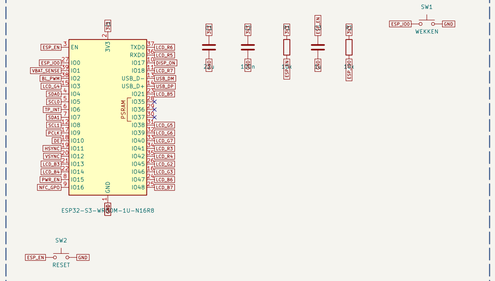
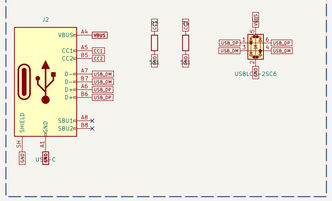
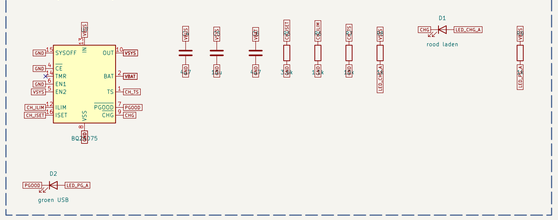
ESP32-S3 (U.FL ANTENNE)



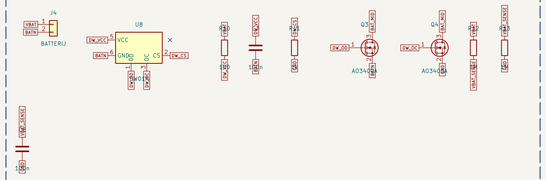
USB-C



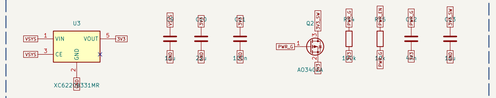
LADEN (BQ24075)



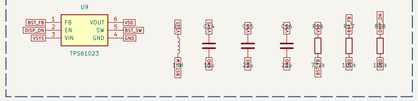
BATTERIJ + BEVEILIGING



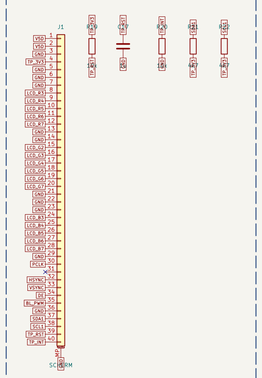
3,3 V + SENSORVOEDING



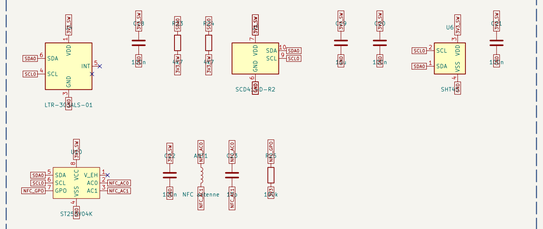
SCHERM 5 V (TPS61023)



SCHERM 4.3" (FFC)



SENSOREN + NFC (I2C0)



SONDE (I2C1)



TESTPUNTEN



BEVESTIGING



LUCHTMETER S3 v2.0 — ESP32-S3-WROOM-1U (external antenna) behind a Waveshare 4.3in 800x480 IPS capacitive touchscreen (generated from tools/design.py)
Screen FFC J1 (40p 0.5 mm, RGB565): physical pin k = screen pin 41−k. R3–7 IO41 IO42 IO44 IO43 IO18. G2–7 IO45 IO46 IO3 IO38 IO39 IO40. B3–7 IO13 IO14 IO21 IO47 IO48. PCLK IO9 DE IO10 HSYNC IO11 VSYNC IO12, backlight PWM IO2.
Touch GT911 on I2C1 (SDA IO7, SCL IO8, INT IO6, address 0x5D). Sensors + NFC on I2C0 (SDA IO4, SCL IO5): SCD41 0x62, SHT45 0x44, LTR-303 0x29, ST25DV 0x53/0x57/0x2D.
Power: USB-C → BQ24075 (charge 250 mA, power path) → LiPo on BAT+/BAT− (DW01A) → XC6220 3.3 V. Screen 5 V from TPS61023 (DISP_ON IO17 high). Sensors + NFC on 3V3_SW (PWR_EN IO15 low = on).
Wake from deep sleep: WEKKEN button (IO0) or NFC field (ST25DV GPO → IO16). Battery voltage = ADC(IO1) x 2.

ESP32-S3-WROOM-1U + 4.3in touch + SCD41 + SHT45 + LTR-303 + NFC, LiPo
Kobe Vanrobaeys & varol56 — maakleerplek
Sheet: /
File: luchtmeter.kicad_sch
Title: Luchtmeter S3
Size: A1 | Date: 2026-10-18 | Rev: 1.0
KiCad E.D.A. 10.0.7 | Id: 1/1